Journal of Humanities and Social Sciences Invention
Vol. 0, No. 0, 2023, 1-7
ISSN: 2581-8791

Environmental Degradation and Sociocultural Survival of Tribes in India

Harshita

Department of Political Science,

Publication Info

Article History:
Received: 00-00-2022
Accepted: 00-00-2022
DOI: 10.35210/jhssi.1.1.1

Keywords:
Social exclusion, Particularly vulnerable tribal groups, Environmental degradation, Social ecology, Environmental exploitation, Sociocultural development.

Corresponding Author:
Harshita
M: +00 000000000
Email: harshitaprajapati9612@gmail.com

Abstract

Environmental degradation is often analyzed at the physical level; however, there is a sociocultural dimension to it as well. Due to unequal distribution of resources, it is tribal communities which bear its cost, so much that some of these tribal communities are either endangered or on the verge of extinction. Environmental degradation thus transcends to the sociocultural and economic realm, putting these communities in a vulnerable position.

INTRODUCTION

The relationship between humans and the environment forms an important part of the larger sociocultural context of any society, and in fact, the latter is shaped to a larger extent by the former. However, this sociocultural value of the environment is undermined and relegated in the analysis of environmental degradation and often is limited to accounting for the physical impact (loss of flora and fauna, depletion of resources etc.) mechanically. This loss due to environmental degradation cannot be simply reduced to "loss of resources" as it also entails subtle but graver mental and sociocultural loss. Furthermore, focus in modern times has been laid on analyzing the impact of sociocultural factors (religion, poverty, population etc.) on the environment rather than another way around

This article endeavors to account for this multi-dimensional impact of environmental degradation by going beyond the materialistic aspect of it. While simultaneously stressing the need to could go beyond institutional and structural responses to counter modern environmental challenges.

The article is divided into five sections. While section one basically introduces the idea of the "rupturing of sociocultural bonds" due to environmental degradation. Sections two and three provide a context of this process while simultaneously juxtaposing the idea of "mainstream" and "primitive" and thus furthering the understanding of the dynamics of environmental degradation. Section four and five seeks to problematize the one-dimensional idea of "development," which makes the condition of tribes deplorable and rips them off any means to participate in society. The central idea thus is that environmental degradation not only implies the depletion of natural resources but also "social exclusion" of those sections of society which are dependent on nature for their survival, both sociocultural and economically.

Environmental degradation: rupturing of cultural bonds:

Environmental degradation's socio-political underpinnings and consequences are highlighted by an eminent American political philosopher, Murray Bookchin. In his essay- What is Social Ecology? he "points out that: "...economic, ethnic, cultural, and gender conflicts, among many others, lie at the core of the most serious ecological dislocations we face today [and] the hierarchical mentality and class relationships that so thoroughly permeate society give rise to the very idea of dominating the natural world." He thus critiques the overall rhetoric of development that lay at the base of capitalism and through which migration and displacement are justified. This class dynamics also highlights the core issue at the heart of every ecological conflict- the issue of what is mainstream and what is not, what development is and what is not. There is an inherent link between the social structures of domination and environmental degradation.

Despite being an independent entity, nature has values and meanings attached to it by people. Further, with the advent of capitalism and scientific development, nature has been commodified and thus, contemporary ecological struggles are between development and "Mati" [2]. Here non-recognition of the "dialectic relationship" between nature and humans is either undervalued or totally ignored and consequently, the rights of minority Adivasis are taken away in the guise of achieving larger and more "important national interests," leading to what Vandana Siva Calls "the ecological and cultural rupture of bonds with nature, and within society".

[2]Through the notion of "desacralization" of soil, Shiva lays emphasis on the process of commodification of "Mati" which has been reduced to the status of "means of production," thus ripping off from it all the cultural, social, psychological and emotional investments that the Adivasis have made in it.

The Continuum of Environmental Exploitation:

This rupturing of sociocultural bonds as a consequence of environmental exploitation and degradation is not taken much seriously. However, scholars like Irawati Karve and Mahasweta Devi have tried to highlight this issue in subtle ways by not only documenting the adversities that the indigenous people face but also by providing an alternative interpretation of texts like Mahabharata and Ramayana from the subaltern point of view, drawing parallels to the capitalist ideology and belief system which subordinates the interests of hunter-gatherers to the strategic need for land and capital as well as socio-political control over the native food gatherers.

The episode of Ekalavya, the incident of the burning of the tribals in the lac house in place of the Pandavas, and the numerous other references to the suppression of tribal peoples are now seen as embodying these early struggles between the tribal and non-tribal populations of India in those early days, for the control over the land and its resources (Gadgil and Guha 78-80). What is novel in their approach is that their analysis goes beyond the physical damage to the sociocultural, emotional and mental impact of environmental exploitation.

So, while on one hand Mahasweta's "Little Ones" provides an account of exploitation that the tribals who are "not as well developed or tall as Russians, Canadians or Americans….(who) has never eaten the kind of calorie-rich food required for the development of a strong human body, the failure to consume which is construed as a crime by World Health Organization" face. On the other hand, in "the book of the hunter" she connects history, myth and her own personal experience with the Shabar tribe, and not only provides an account of the proximity between the tribals and nature but also the atrocities that they must face due to the constant expansion of city area. Throughout the story, there is a constant tussle between the ideology and customs of the town dwellers and the tribals. Both of literary works probe into two different dimensions of environmental degradation and exploitation- while the former is more about the physical aspect the latter is about the sociocultural aspect.

Going beyond Vedic times, scholars like Ramchandra Guha and Madhav Gadgil provided a detailed account of ecological exploitation and oppression met by the tribals at the hands of colonizers, in some cases even leading to extinction. The forest policies adopted by the colonial government like making hunting activities illegal, commercialization of forests and forceful change in cultivation techniques, led to the sharp decline in the population of the tribal communities.

The Chenchus of Hyderabad for instance, were deeply impacted by the commercial forestry and opening of their area to frequent contact with non-tribal. They were relegated to the status of "agrestic serfdom".

Similarly, the Birhor tribe of Chotanagpur, Baigas of central India, witnessed a sharp fall in the population, the former declining –from 2340 in 1911 to 1610 in 1921. The disfavor of traditional shifting cultivation or jhum as "the most destructive of all practices for the forest" had further implications as tribal life was structured on it. It was not merely an economic and agricultural practice rather, it was part of a larger sociocultural construct. The change that one notices here is that now the forest is not just the source of subsistence but has become a symbol of culture and pride. A linkage is developed between the forest and the people, which provides them with history and unique cultural identity[3]. While simultaneously, it underlines the perishing effect that the ban on subsistence farming had on their culture since their link to nature broke. Even the artisanal classes, particularly the bamboo industry as well as pastoral nomads, were badly impacted by the colonial environmental policy.

These examples thus not only highlight the impact of environmental policies at the physical level, i.e., at the economic level, and the mental and social impact of such policies is also hinted at. They hint at the graver consequence of environmental exploitation- i.e., negative changes in the social fabric of society.

Post-colonial era and multi-dimensional environmental exploitation:

However, even after the end of colonial rule, these indigenous Adivasis' condition remains unchanged despite various constitutional provisions. To understand this continuation of ideology, it is important to analyze the glaring difference between what has been ensured by the Constitution and what is the ground reality of post-colonial times. Adivasis constitute 8.6% of India's population and are particularly prominent in Andhra Pradesh, Chhattisgarh, Gujarat, Jharkhand, Madhya Pradesh, Maharashtra, Odisha, West Bengal and some northeastern states, and the Andaman and Nicobar Islands continue to face grave socio- cultural circumstances due to environmental degradation.

[3] Vinita Damodaran provides an example of such linkage in the context of Chotanagpur where the festivals of the sacred grove were particularly important for the settled agricultural communities of Chotanagpur and emphasized the importance of the forest and its flowering seasons in the ritual life of the communities. The forest was a sphere where the spiritual and moral spheres of human life intermingled. A symbiotic relationship thus developed between the people and the landscape in Chotanagpur.

This is even though Article 46 of the Indian Constitution, the Protection of Civil Rights Act 1955, Scheduled Castes, and Scheduled Tribes (Prevention of Atrocities) Act 1989, Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Rules 1995 as well as The National Commission on Scheduled Castes and Scheduled Tribes, have been established as a safeguard mechanism. An analysis of environmental crimes and crimes against STs provides an alternative picture of social justice and reinstates the argument that India's environmental policy still looms in the institutional and structural framework, with hardly any alternative being looked at from the sociocultural perspective.

These statistics point to an interesting observation. The states which have high rates of crimes against Scheduled tribes and environmental crimes are those that house diverse indigenous tribes. And these crimes basically take cognizance of the physical loss to the environment and tribes. The sociocultural impact of environmental degradation is totally unaccounted for. This can be supported by the fact that in the Andaman and Nicobar Islands, where the tribes are on the verge of extinction, cases reported against environmental crimes are nil.

'Us' and 'Them'- missing 'social' in 'environment'

As a result of this oppression and apathy on the part of non-tribals, India is losing its diverse indigenous cultures, a recent example of which is the Andamanese tribe Bo, which has got extinct in 2010. The factor here in the place was the tsunami and

Table 1: Crime/Atrocities against Scheduled Tribes(STs) – 2014-2016

S. No	*States*	*2014*	*2015*	*2016*	*Rank Based on Incidence/ %share (2016)*
1	Rajasthan	129	101	88	1
2	Hyderabad	8	16	21	2
3	Gujarat	8	4	10	3
4	Karnataka	11	12	10	4

Source: National Crime Records Bureau, Crime in India (2016) Statistics

Table 2: Environment Related Offences 2014-2016

S. No.	*State/UT*	*2014*	*2015*	*2016*	*Rate of Total Cognizable Crimes (2016)++*
1	Uttar Pradesh	1597	1779	2130	1.0
2	Rajasthan	2927	2074	1381	1.9
3	Karnataka	249	211	178	0.3
4	Jharkhand	148	233	139	0.4

Source: National Crime Records Bureau, Crime in India (2016) Statistics

the migration of fellow tribals to urban, developed cities. Before Bo senior in 2004, her friend Boro died. This left her secluded socially because no one else shared her language. According to Dr. Anvita Abbi, from the total population of 5,000 only 50 Great Andamanese are left. With the death of Bo senior or Boa, the last member of the tribe, India has lost its two indigenous languages- Bo and Khora, and we now have only Jeru and Sare speakers in the group. This is the most subtle form of exploitation and harassment caused by environmental degradation, whereby one cannot even express one's feelings due to the loss of one's language. It further leads to the social seclusion of already deprived tribals. This sense of loss of one's culture has far more psychological impact, which we fail to assess and account for. Overall, all five tribes in Andaman Nicobar Islands are on the verge of extinction.

The aboriginal tribes of Attappady, with their rich culture and tradition that go beyond the Vedic times, seem doomed to extinction, afflicted with major health problems like infertility, sickle cell anemia, tuberculosis and thrombi angiitis obliterans. The Asur Adivasis, credited with being the country's first community of iron smelters, are today one of India's dwindling, particularly vulnerable tribal groups and one of the smallest Adivasi groups in Jharkhand. Asuri, their language, is on the verge of extinction, with less than 8,000 people speaking it. New mining technologies and forest laws that restrict the forest-dwelling Asur from burning wood to produce the charcoal used in their smelting work have slowly made their traditional means of livelihood redundant. For the last eight years, the 200-odd Asur Adivasi families living in the "villages of Ghorapahad, Sekuapani and Amtipani in Gumla have worked at Hindalco's bauxite mines". The Asur inventors, the kings of the forest kingdom, have become daily-wage workers in their ancestral land.

The languages or dialects which are considered endangered, include 11 from Andaman and Nicobar Islands (Great Andamanese, Jarawa, Lamongse, Luro, Muot, Onge, Pu, Sanenyo, Sentilese, Shompen and Takahanyilang), seven from Manipur (Aimol, Aka, Koiren, Lamgang, Langrong, Purum and Tarao) and four from Himachal Pradesh (Baghati, Handuri, Pangvali and Sirmaudi). The other languages in the endangered category are Manda, Parji and Pengo (Odisha), Koraga and Kuruba (Karnataka), Gadaba and Naiki (AP), Kota and Toda (Tamil Nadu), Mra and Na (Arunachal Pradesh), Tai Nora and Tai Rong (Assam), Bangani (Uttarakhand), Birhor (Jharkhand), Nihali (Maharashtra), Ruga (Meghalaya) and Toto (West Bengal). Nearly 400 of India's 850-odd languages face the threat of extinction because of an erosion of traditional jobs that are fueling migration to cities.

Ganesh Devy, founder and director of the Bhasha Research and Publication Centre in Vadodara and the Adivasi Academy in Tejgadh, Gujarat, also argue that many coastal, nomadic and Himalayan languages could vanish within half a century as "we have lost 250 languages since 1961 and another 400 will perish in the next 50 years. The languages spoken in the coastal areas will be the worst-affected". The large-scale mechanization of coastal activities like fishing and salt production had forced local people to migrate hordes, endangering their connection with their native culture.

The environmental degradation thus also makes social participation of such tribes intractable by stratifying the "Adivasis" as "others" eventually leading to continuous degradation and depletion of their culture.

Devy argues that "in their ethnic, cultural and linguistic attributes, th[ese tribes] are so varied that it is almost impossible to speak of them as a common category of humanity. No single term can describe them with any degree of semantic assuredness. Even if one were to accept one or another term for the purpose, its normative frame may run up against numerous contradictions with the strikingly divergent history of every community…… an abdication of anthropology which argues that the tribe can survive only by becoming the "Other" [4]. This also leads to overgeneralization and simplification of sociocultural context[5].

[4](i) The beginning of this process could be traced to the creation of "scheduled districts" in India by the British colonial government during 1874, which demarcated and separated the socio-political practices of tribes and colonial administration. As a result, there is a progressive construction of tribal identity as Aranyakas, Melecchas, primitive, and indigenous which is juxtaposed to "mainstream society" in a romanticized way.

(ii)Even the United Nations working group established in 1982 for determining the communities that can be described as indigenous, came up with four-fold criteria: (a) "pre-existent" peoples; (b)

"marginalized and dominated" people; (c) "minority or culturally different" people; and (d) people who identify themselves as indigenous, thus socially constructing them as "others".

The process of transcendence of this socially constructed "other" to a political one is characterized by Baviskar as "protection vs. assimilation" with Verrier Elwin and G.

S. Ghurye as the respective proponents of these opposing positions. While Elwin advocated special provisions for protecting the rights of the tribal population, sighting sociocultural and economic differences between tribals and non-tribals. Ghurye on the other hand refuted this argument, believing that "Backward Hindus" were akin to the Hindus6 to the extent that there were no differences between them, which should be addressed through special provisions.

Environmental Barriers and Politics of Development

Environmental degradation in tandem with sociocultural loss as well as "economic and political impoverishment has prevented most tribal communities from negotiating a better deal[7] for themselves vis-a-vis displacement and development." Already disadvantaged by the processes of state-led industrialization, many tribal groups have scarce material and "symbolic capital" [8] to lean on. This leads to their interests getting submerged by the meta-narrative of "national interest" and continued practice of the dominant section defining the social structures and rights of the minority section according to their whims and fancies. The "inclusion" and "mainstreaming" of these "primitive tribes" is also dictated by these dominant sections of society.

[5]Amita Baviskar, highlights this overlooking and generalization of sociocultural context, through the instance of data collection method used for the Sardar Sarovar Project, whereby "The bulk of the report consisted of general data about Bhils culled mainly from government inventories of the manners and customs of the natives"…. homogeniz(ing) the reality of Narmada itself. The "Bhil and Bhilala people living in the plains of Nimar in Madhya Pradesh have been Hinduized and absorbed into caste-based village society in a way that does not apply to the Bhils, Bhilalas and Tadvis of the hills of Maharashtra, Gujarat and Jhabua (Madhya Pradesh) in the submergence zone" (Baviskar 1995:83-5). Thus, lacking in any kind of sensitivity to a broader social and cultural realm in which the Adivasis reside.

[6] The concept of Hinduized Tribals has its underpinnings in the notion of "Hindu" as a homogenous theological religion, through which a uniform cultural identity is ascribed to diverse cultures which are present in the society. This practice could be traced back to the introduction of the census by the British government in the colonial period.

[7] Sumit Guha provides an excellent analysis of this aspect in the context of Dang and the kokni/kunbi tribes whereas the ability to obtain state resources or to occupy and retain forest land depended crucially on the ability to utilize fora such as cooperatives and the electoral system. Opportunities in the modern political and economic system were therefore best utilized by communities that were already moving towards peasant farming.

[8] In sociology and anthropology, symbolic capital can be referred to as the resources available to an individual on the basis of honour, prestige or recognition, and serves as a value that one holds within a culture. A war hero, for example, may have symbolic capital in the context of running for political office.

The continuous exploitation of the Jarwa tribe has been facing and the Supreme Court verdict that followed up after a case was filed for seeking redress exemplifies how institutionalized efforts of "social inclusion" can lead to further exclusion. This tribe is now on the verge of extinction due to various factors including poaching, excessive tourism due to the opening of Andaman Trunk Road, and abuse at the hands of the tourists, since due to tourism they have come in contact with diseases like measles, which is proving dangerous for their health.

It was to address these issues that Supreme Court in the case Lieutenant General Andaman and Nicobar Islands and Ors vs. M/S Barefoot Inns and Leisure Pvt. Ltd, under section 3(1) of the Andaman and Nicobar Islands (Protection of Aboriginal Tribes) Regulation, 1956 declared an area up to a five km radius around the Jarwa Tribal reserve as a buffer zone. Entry of any person except the aboriginal tribe in the buffer zone was dismissed. Further, the Court declared that "the administration or any private individual shall carry out no commercial and tourism-related activities…." However, in 2013 the Court changed its decision and allowed the profitable Human Safaris to continue. Survival International, an NGO working for tribal rights came out with a report- "Progress Can Kill", which brought forward the ordeals that indigenous tribes have to face in the wake of so-called "development" which was being imposed on them. Such a forced and one-sided idea of development has affected their health to an unimaginable degree.

Another way in which the rights of Adivasis are being hampered is under the garb of conservationism. According to present conservation politics, about 600.000 people have already been displaced out of India's protected zones. As a result of the government's investigations, at least 75% of these communities have not been relocated conveniently. Global assessment of displacement from national parks in rainforest areas concluded that millions of "conservation refugees" have been displaced, or are facing displacement risks within the next few years (Geisler 2003). As Maria Costanza Torri argues, "Displacement of local communities out of protected zones can entail their pauperization and disappearance of traditional culture that links them with wildlife and forest biodiversity".

What both these cases underline is the issue of agency[9]- the agency of tribes in dealing with state-sponsored policies as well as with various actors from civil society to state actors.

[9]The seed cultivation and farming techniques of Dongria Kondh of Niyamgiri Hills exemplify the positive impact that could be created by giving tribes "agency" and non-interference. This tribe through their elaborate cultural rituals and agricultural practice involving farmer, bejuni (senior village priest), and barik (messenger) have successfully been able to revive Kodo millet- a variety high in fibre and energy content after an absence of 40 years.

The unilateral adjustment from the side of the excluded has denied these tribes their social history. They have become non-participants in their own inclusion.

The Pathalgadi movement which has spread in the states of Jharkhand, Chattisgarh and Orissa serves as a pertinent example to understand the politics of development as well as of socio-political participation in the context of post-colonial governmentality, by highlighting the sociocultural as well as political underpinnings of the "democratic assertion" of the "self" by the tribes of the scheduled districts. The custom of Pathalgadi, originally called Sansandiri, entails the erection of stone at the tomb of a dead person and is primarily practiced by tribes belonging to the Austro- Asiatic linguistic families such as –the Mundas, Khasis etc. The movement thus symbolizes the transcendence of "traditional" to the "political", whereby traditional cultural symbols are used to organize popular movements in defiance of the state and the machinery that imposes it (Singh, 2019). In response and defense against the exploitative practices of the state: the amendment of the Chotanagpur Tenancy Act(1908) and the Santhal Pargana Tenancy Act (1876), as well as Land Acquisition, Rehabilitation and Settlement(Jharkhand Amendment Bill) 2017, and introduction of Jharkhand Freedom of Religion Bill(2017) along with Momentum Jharkhand Summit in 2017; the movement was

initiated in a small nondescript village named Bhandra of Khunti district in Jharkhand, by erecting a stone slab on the boundary of the village on 9 March 2017 (Hindustan Times 2018).

The Pathalgadi movement not only highlights the importance of linkage between the customary practice of "parha panchayat", the traditional panchayats of the tribes of Chotanagpur for asserting the self-identity as well as the autonomy of tribes in fifth scheduled areas since it was a sacred institution as claimed by the famous Munda saying –"Simare Singbonga Otere Panch (the Sun god above and the Panch on earth) (Roy, 1970;66), but it also indicates towards the way in which these tribes interpret the constitutional provisions within their sociocultural understanding while simultaneously interpreting and asserting their identity.

CONCLUSION

As Ramachandra Guha argues that "the contemporary ecological problems cannot be properly understood without relating it to the problems within society as "the biophysical and sociocultural domains" are "interdependent". It becomes extremely necessary to analyze the sociocultural cost of environmental degradation before putting at stakes the interest of subaltern groups at "the alter of development" because their life isn't only dependent but also shaped by their natural surroundings. It is not the tribals who are hampering our right to development, rather it is the non-tribals who are

exploiting them and making them further vulnerable.

The environmental discourse should seek to achieve a balance between the "isolationist" approach as advocated by Elwin on the one hand and the integrating and assimilative approach on the other. As historical experience shows, it could be "ethnocide" as it exposes the tribes to diseases to which they had no resistance – leading to vast casualties and decimation of the tribes. The latter point has been recognized even by the Bhuria Committee. Whereby they found "that the tribes have reservations about it, as being not only paternalistic and patronizing but also prejudicial to the indigenous and tribal peoples' ethnic identity and more, and violative their rights. They are averse to overt or covert attempts that aim at their assimilation. They wish to preserve the integrity of their culture and personality through the first of the five principles of the Panch Sheel of Tribal Development, which exhorts that the tribal people should be allowed to develop along the lines of their own genius." They have to be an active agent in their own inclusion.

REFERENCES

Shiva V, and Maria Mies (2014) Ecofeminism, London: Zed Books

Saravanan V, - Environment History and Tribals in Modern India (2018), Palgrave Macmillan

Baviskar A, (1995) The Political Uses of Sociology: Tribes and The Sardar Sarovar Project, Sociological Bulletin, volume 44, pp- 89-96

Torri M, (2011) Conservation, Relocation and The Social Consequences of Conservation Policies in Protected Areas: Case Study of the Sariska Tiger Reserve India, conservation and society 9(1), pp – 56-64

C. A, Bayly and Gordon Johnson (eds)(1999) Environment and Ethnicity in India pp1200-1991, (Cambridge Studies in Indian History and Society) New York, Cambridge University Press

Hainendorf, Christoph Von (1982) Tribes of India the Struggle for Survival, University of California Press

Damodaran V (1997) Environment Ethnicity and History in Chotanagpur Environment and History volume 3 pp- 273-298

Devi Mahasweta(1998) Bitter Soil, translated by, Ipsita Chanda, Calcutta, Seagull Books

Devi Mahasweta(1999) The Book of The Hunter translated by Mandira and Sagree Sen Gupta, Calcutta Seagull books

Pal, Mahi. "Panchayats in Fifth Scheduled Areas." Ecoomic Political Weekly May 2000: 1602-1606.

"Politics of Language, Religion." Economic Political Weekly March 2005: 1363-1370.

Singh, Anjana. "Many Faces of the Pathalgadi Movement in Jharkhand." Economic Political Weekly MARCH 2019: 28-33.

Sundar, Nandini. ""Custom" and "Democracy" in Jharkhand." Economic and Political Weekly October 2005: 4430-4434.

https://www.indiaseminar.com/2009/601/601_ganesh_n_devy.htm

https://www.telegraphindia.com/india/extinction-alert-on-400-languages/cid/1521283

https://www.thehindu.com/todayspaper/tpnational/tpkerala/Attappady- tribals-face-threat-of-extinction/article14890483.ece

https://scroll.in/roving/827128/the-asur-adivasis-indias-first-metallurgists- now-struggle-for-daily-wages-in-jharkhands-mines

https://indiantribalheritage.org/?page_id=1000

Query Report

Q1 Kindly provide corresponding author contact number.

Q2 Kindly provide Short title of the article.

Q3 Kinly provide author full affiliation.

Q4 Kindly provide table 1 and 2 in-text citation

Q5 Kindly provide reference in-text citations in APA format.

Q6 Cross-check for accuracy/data/format, correct, and provide all references in consistent APA format. Sample reference formatted to act as an example and to maintain consistency of format (APA) required

Chandra, S., Sikdar, P. K., & Kumar, V. (1995). Dynamic PCU estimation of capacity on urban roads. Journal of Indian Roads Congress, 23, 17–28.

Chandra, S. (2004). Capacity estimation procedure for two-lane roads under mixed traffic conditions. Journal of Indian Roads Congress, 165(1), 139–170."